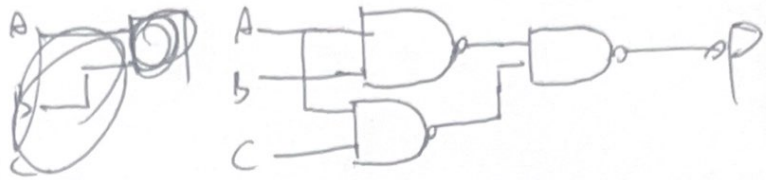
2-8.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$F = AB + AC = \overline{\overline{AB}\ \overline{AC}}$



2-11. (5). $F_5 = \overline{(\overline{AB})}\,(\overline{\overline{A}CD})\,(B+\overline{C}+\overline{D})$

$= (\overline{A}+\overline{B})(A+\overline{C}+\overline{D})(B+\overline{C}+\overline{D})$

$= (\overline{A}\overline{C} + AB + \cancel{A\overline{C}} + \overline{A}\overline{D} + \cancel{A\overline{D}})(B+\overline{C}+\overline{D})$

$= \overline{A}B\overline{C} + \overline{A}\overline{C} + \overline{A}\overline{C}\overline{D} + AB + AB\overline{C} + AB\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}\overline{C}\overline{D} + \overline{A}\overline{D}$

$= \overline{A}\overline{C} + AB + \overline{A}\overline{D}$

2-12. (1).

AB \ C	00	01	11	10
00			1	
01			1	
11	1	1	1	1

$F_1 = AC + AB + BC$

(4).

AB \ CD	00	01	11	10
00		1	1	
01	1	1		
11		1	1	
10	1	1		

$F_4 = \overline{A}\overline{B}D + \overline{A}B\overline{C} + AB\overline{D} + A\overline{B}\overline{C}$

2-13. (3). $AB+AC=0$

$\bar{F} = \bar{A}\bar{B}\bar{C} + \bar{A}B + A\bar{B}\bar{C}\bar{D}$

$= \bar{A}\bar{B}\bar{C} + \bar{A}B + AB + A\bar{B}\bar{C}\bar{D}$

$= \bar{A}\bar{B}\bar{C} + B + A\bar{B}\bar{C}\bar{D}$

$= \bar{A}\bar{B}\bar{C} + B + A\bar{C}\bar{D} + AC$.

$= \bar{A}\bar{C} + B + A\bar{C}\bar{D} + AC$

$= B + C + A\bar{C}\bar{D}$.

$= B + C + (A\bar{D})$.

2-19. (1)

AB \ CD	00	01	11	10
00	1	1	1	
01		1		
11	X	X	X	X
10	1		X	X

$\bar{F} = \bar{B}\bar{C}\bar{D} + \bar{A}\bar{C}\bar{D} + (\bar{A}\bar{B}\bar{D})$

$\bar{F} = (\bar{B}\bar{C}\bar{D}) + (\bar{A}\bar{C}\bar{D}) + (\bar{A}\bar{B}\bar{D})$

(6).

AB \ CD	00	01	11	10
00	1			1
01		X	X	1
11		X	X	1
10	1			1

$F_6 = B\bar{D} + C\bar{D}$.